

Google Play Store Apps Analysis



Google Play Store

By – Rishav Sinha



Introduction

- **Objective:** Analyze Play Store dataset to extract insights about app performance, categories, ratings, and user reviews.
- **Datasets:**
 - 1- **Playstore_df** → App details (category, size, installs, price, ratings)
 - 2- **User_df** → User reviews with sentiment labels.

Data Overview

- **Playstore_df** → Number of rows, columns, missing values.
- **User_df** → Number of reviews, missing values.
- **Issues found:**
 - Null values in ratings, size
 - Non-numeric characters in numeric fields

Data Cleaning

- **Converted installs, size, and price to numeric format.**
- **Replaced invalid entries with NaN.**
- **Standardized size (KB, MB → MB)**
- **Dropped irrelevant or junk apps.**

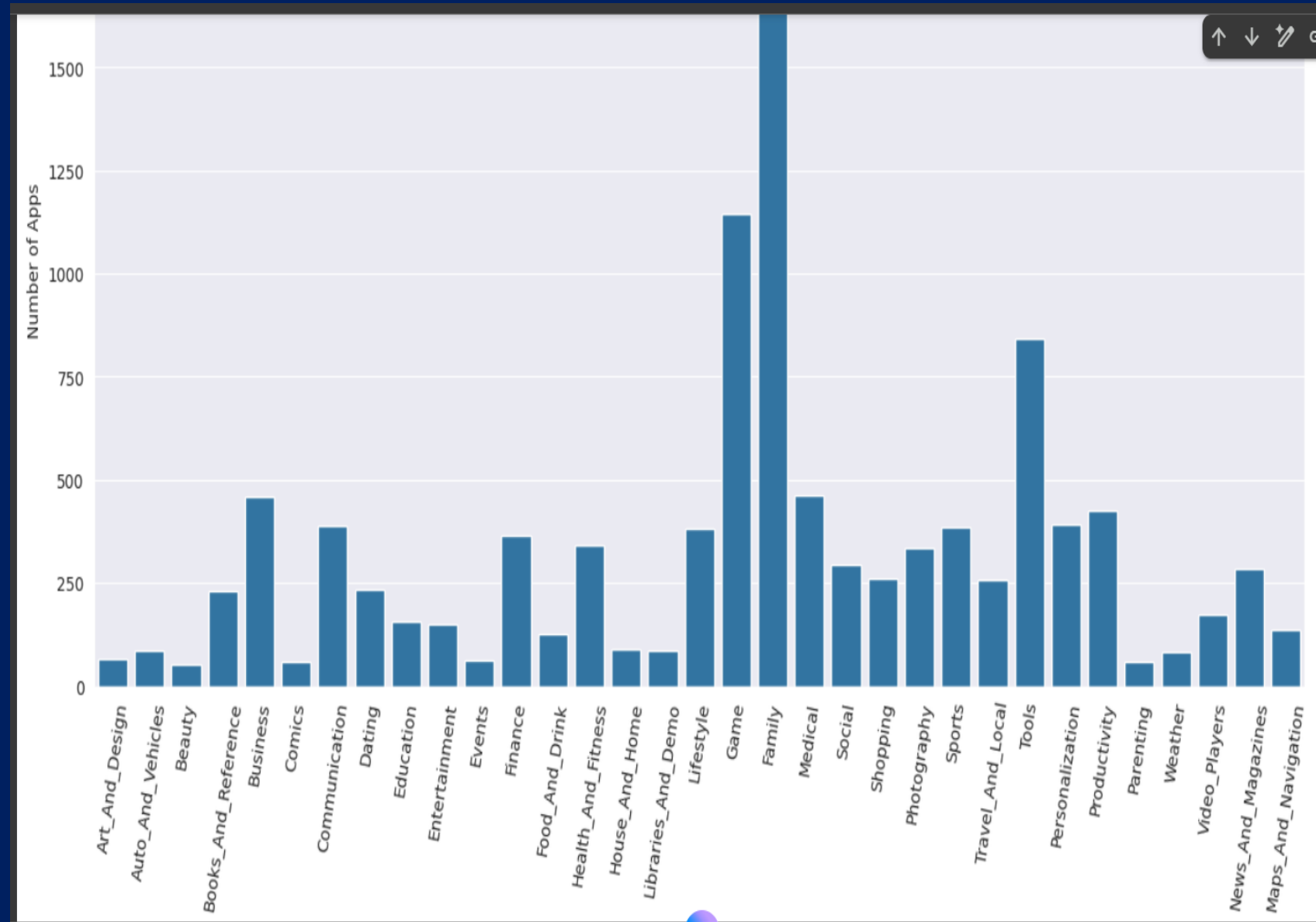
App Ratings Analysis

- **Distribution of ratings.**
- **Top 10 highest-rated apps by category.**
- **Insight: Certain categories consistently maintain higher user ratings.**

	App	Category	Rating
10776	Monster Ride Pro	Game	5.0
615	Awake Dating	Dating	5.0
612	American Girls Mobile Numbers	Dating	5.0
654	Speeding Joyride & Car Meet App	Dating	5.0
10820	Fr. Daoud Lamei	Family	5.0
10529	FK CLASSIC FOR YOU	Business	5.0
10416	FH School	Family	5.0
10407	Jigsaw Volvo FH 16 Trucks	Family	5.0
10357	Ríos de Fe	Lifestyle	5.0
1547	Eternal life	Libraries_And_Demo	5.0
2271	FHR 5-Tier 2.0	Medical	5.0

Category-Wise Analysis

- Number of apps in each category (bar chart)
- Categories with most apps: (e.g., Family, Tools, Productivity)
- Categories with fewer apps = potential opportunities



Downloads & Popularity

- Top 10 apps with most downloads (table)
- High downloads ≠ High ratings (insight)
- Example: Some apps have millions of downloads but low user satisfaction

	App	Installs
2808	Google Photos	1.000000e+09
2853	Google Photos	1.000000e+09
3223	Maps - Navigate & Explore	1.000000e+09
3117	Maps - Navigate & Explore	1.000000e+09
3234	Google	1.000000e+09
3232	Google Street View	1.000000e+09
3127	Google Street View	1.000000e+09
2554	Google+	1.000000e+09
464	Hangouts	1.000000e+09
468	Skype - free IM & video calls	1.000000e+09
451	Gmail	1.000000e+09

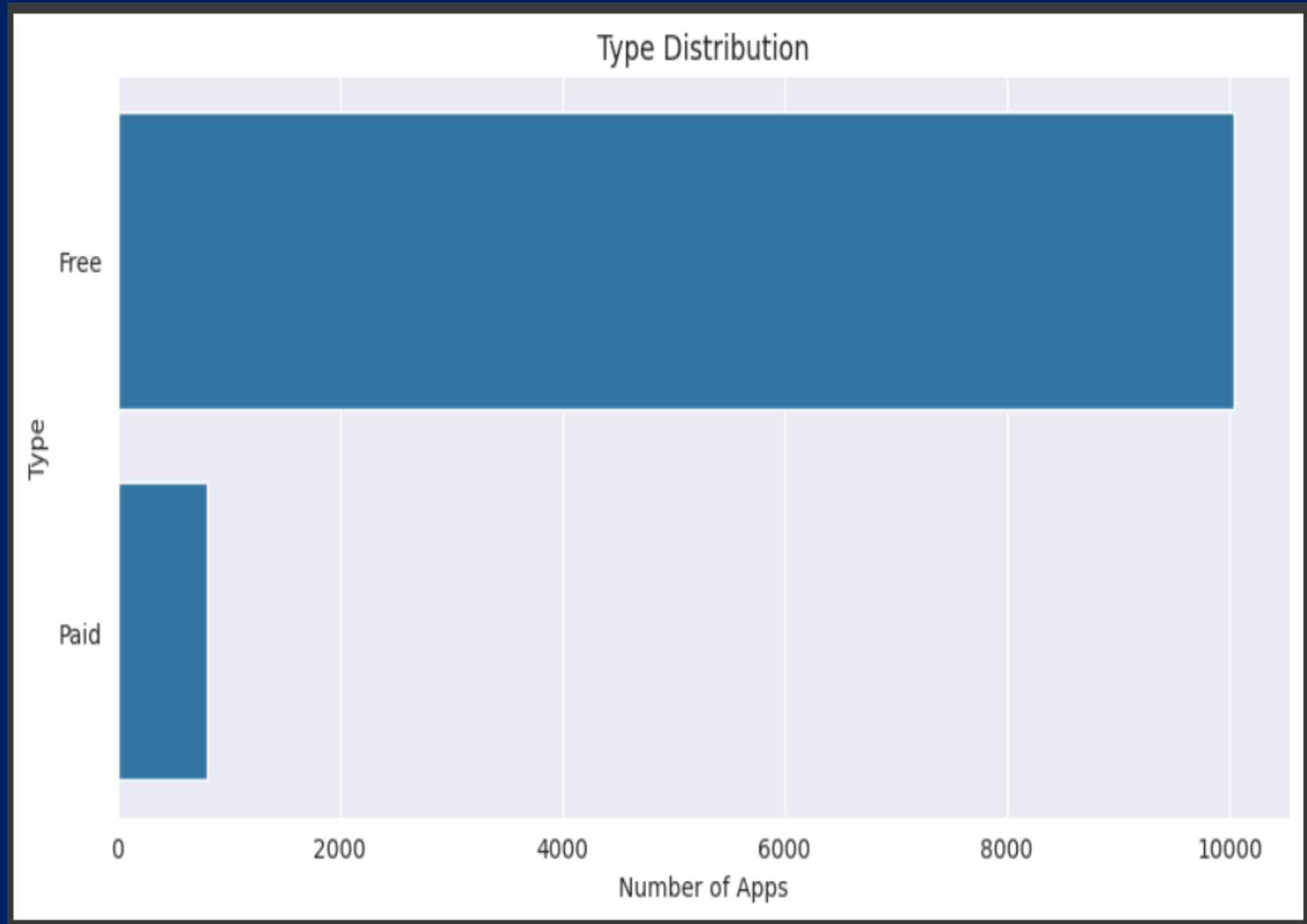
Family Category Insights

- Lowest-rated vs. Highest-rated apps in Family category
- Insight: Family apps dominate the store but vary greatly in quality

	Category	Rating
5978	Family	1.0
10324	Family	1.0
4127	Family	1.0
4905	Family	1.6
6135	Family	1.7
9176	Family	1.7
4904	Family	1.8
6812	Family	1.8
7416	Family	1.9
6532	Family	1.9
9890	Family	1.9

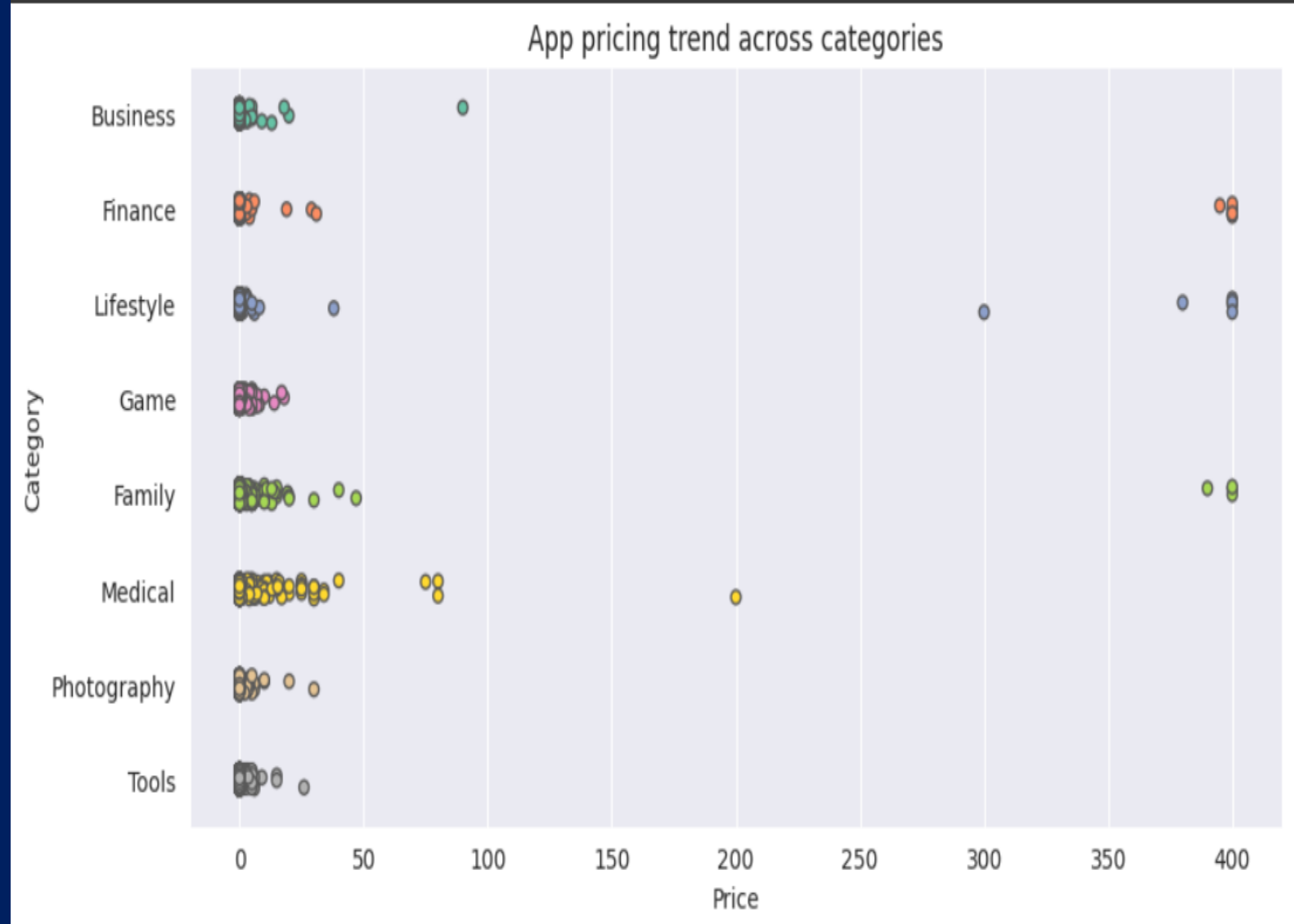
Free vs Paid Apps

- **Comparison: Ratings & downloads of free vs paid apps**
- **Most apps are free → ad-based revenue model dominates**
- **Paid apps tend to have slightly better ratings**



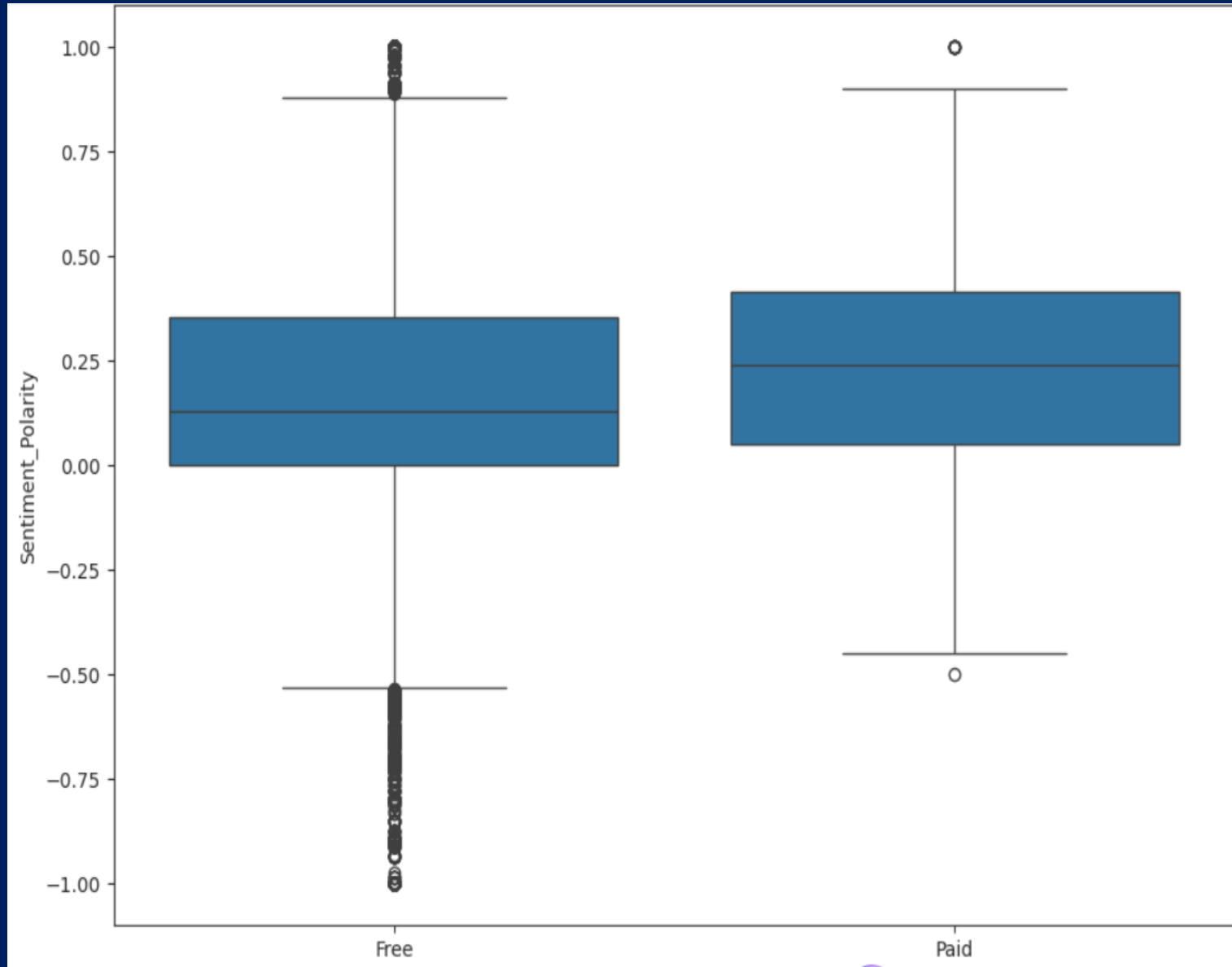
Price vs Category

- **Relation between app price and category**
- **Very few categories have premium apps (Books, Education, Lifestyle)**
- **Free model dominates entertainment categories**



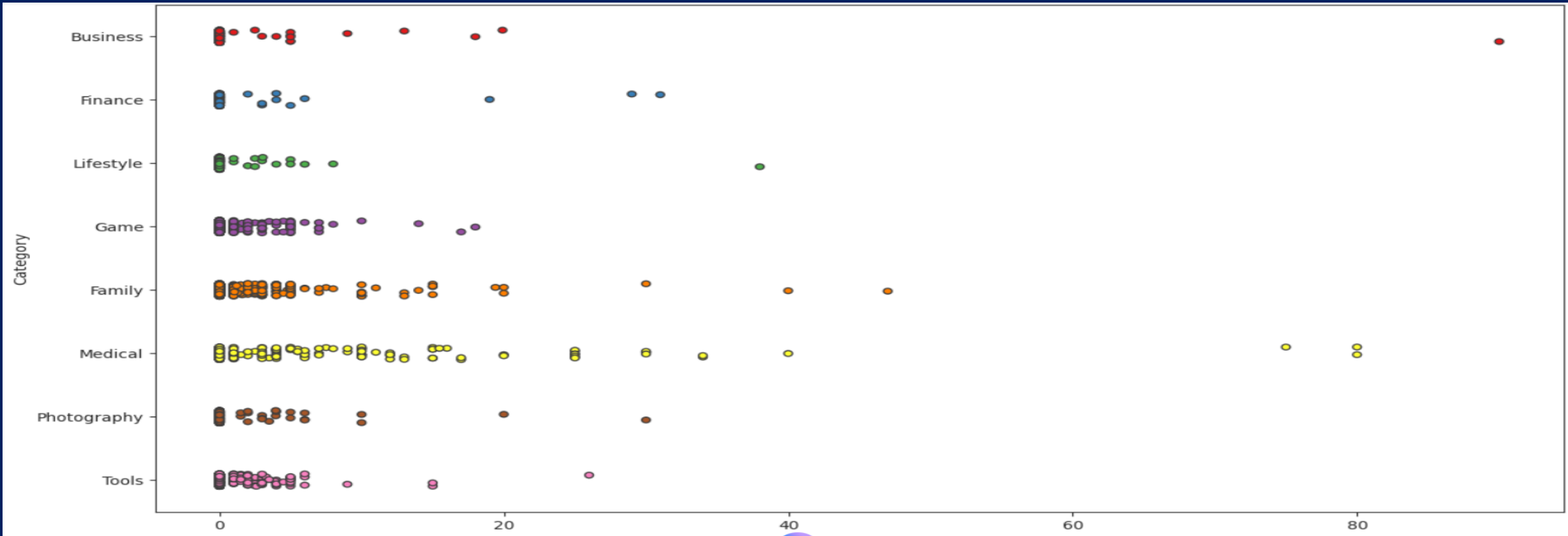
Sentiment Analysis

- Performed sentiment analysis on user reviews
- Categories: Positive, Negative, Neutral
- Insight: Majority of reviews are positive, but negative reviews highlight performance & bug issues



Filtering Out Junk Apps

- Defined criteria for junk apps (e.g., very low ratings, low installs, irrelevant apps)
- Chart: Proportion of junk vs. valid apps
- Insight: Removing junk apps improved dataset quality & analysis accuracy



Business Insights

- **Family category → largest scope, needs quality improvement.**
- **Free apps dominate → opportunities for monetization via ads & freemium models.**
- **Certain categories underserved → new apps can fill gaps.**
- **Sentiment analysis helps in identifying improvement areas.**

Conclusion

- **Cleaned & analyzed Google Play Store dataset.**
- **Identified top categories & apps by rating/downloads.**
- **Found gaps in categories with fewer apps.**
- **Suggested opportunities for growth in Family & premium apps**

Thank You